

Andy Banh

437-230-1602 | andybanhab@gmail.com | linkedin.com/in/andybanh | github.com/AndyyBanh

EDUCATION

York University

Hons. Bachelor of Science in Computer Science

Toronto, Ontario

Expected April 2028

Relevant Courses

Advanced OOP, Data Structures and Algorithms, Theory of Computation, Introduction to Database Systems

EXPERIENCE

Sole Proprietorship

April 2025 – Present

- Founded and managed an ecommerce business, managing full operations including inventory, sales, and customer service, for **500+** products
- Analyzed sales data and trends to optimize product sourcing and pricing, generating over **\$60k** in revenue in 2025

Software Developer Intern

Feb 2026 – April 2026

Noble Solutions – Riipen

Remote

- Contributed to implementation plans and development of real estate platform, earning **5-star feedback** on Riipen
- Engineered a data processing pipeline for **10k+** properties, reducing redundant computations and latency by **90%**
- Designed and deployed automated cron-based workflows, saving clients hours of manual efforts
- Technologies used: Next.js, Python, MongoDB, Supabase

PROJECTS

Financial API Gateway | *Spring Boot, Spring Cloud Gateway, Kafka, Redis, OpenFeign, Mockito, Docker, PostgreSQL*

- Architected an API gateway with Spring Security and Redis rate limiting, enabling independent scaling
- Designed an event-driven processing pipeline using Kafka, achieving average of **492ms** end-to-end latency
- Load tested transaction creation endpoint with **k6 (50 Vus)** hitting 100% success rate across **900+** requests
- Engineered a fraud detection service consuming Kafka events with cross-service Feign lookups, enabling decoupled asynchronous background processing

Facility Management System | *React.js, Spring Boot, Mockito, Redis, Docker, PostgreSQL*

- Built **40** JUnit & Mockito unit tests across the service layer, achieving an **81%** branch coverage
- Implemented caching using Redis to reduce backend load, eliminating **99%** of queries on static reference data
- Engineered a tiered Bucket4j & Redis rate limiter, capping public traffic at **100 req/min** and login at **5 req/min**
- Integrated JWT Authentication with Spring Security, securing endpoints and user data

JPMorganChase Software Engineering Simulation Certification | *Spring Boot, Kafka, H2*

- Constructed a Kafka powered Spring Boot microservice to consume, validate, and persist transactions
- Validated sender and recipient balances before processing, preventing invalid or insufficient fund transfers
- Integrated external Incentive REST API using RestTemplate to automate reward payouts per transaction

Form Coach | *Next.js, Spring Boot, Python, OpenCV, Mediapipe, FastAPI*

- Developed a full-stack computer vision platform for workout form, resulting in real-time automated feedback
- Optimized WebSocket streaming pipeline through image downscaling, reducing payload by **42%**
- Implemented flow control through pending-frame tracking, achieving **10 FPS** at **36ms** average latency

Discrete Math Chatbot | *Next.js, TypeScript, Python, LangChain, Chroma, FastAPI, OpenAI API*

- Built a **RAG** pipeline indexing **873** chunks across Discrete Math chapters allowing for context-aware responses
- Validated **GPT-5.4-mini** response generation across real test queries, averaging **1.4s** latency

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, C, Python, SQL, HTML, CSS

Frameworks & Libraries: Spring Boot, React.js, Next.js, FastAPI, Tailwind CSS, JUnit, Mockito

Databases: PostgreSQL, MongoDB, Redis, H2

Developer Tools: Git, Docker, Kafka, AWS, Linux, Maven, Postman, Supabase, Grafana k6